

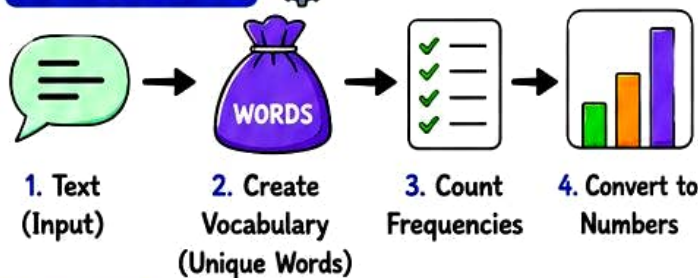
Bag of Words & Polarity Score

Turning text into numbers and understanding feelings behind words!

1 Bag of Words (BoW)

A simple way to convert text into numbers by counting the frequency of each word, ignoring grammar and order.

How it works?



Example

Input Texts	Vocabulary (Unique Words)	BoW Matrix		
		Doc 1	Doc 2	Doc 3
I love natural language processing	i	1	0	1
	love	1	0	1
	natural	1	1	0
Natural language processing is fun	language	1	1	0
	processing	1	1	1
	is	0	1	0
I love text processing	fun	0	1	0
	text	0	0	1

★ Key Points

- ✓ Easy to implement and understand
- ✓ Works well for many machine learning tasks
- ✓ Does not consider word order or meaning

Use Cases



Text Classification



Spam Detection



Sentiment Analysis



Information Retrieval

2 Polarity Score

A measure used in sentiment analysis to determine how **positive**, **negative**, or **neutral** a piece of text is.

How it works?

$$\text{Polarity Score} = \frac{\text{Positive Score} - \text{Negative Score}}{\text{Positive Score} + \text{Negative Score} + 0.000001}$$

Score Range: **-1 (Very Negative)** to **+1 (Very Positive)**

Example



"I love this product. It is amazing and wonderful!"

Positive: 3
Negative: 0
Polarity Score: +1.00



"This is the worst experience. I hate it!"

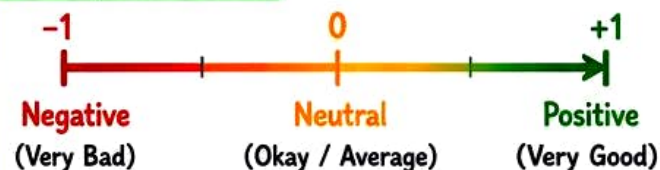
Positive: 0
Negative: 2
Polarity Score: -1.00



"The product is okay, not bad but not great."

Positive: 1
Negative: 1
Polarity Score: 0.00

Polarity Score Guide



★ Key Points

- ✓ Helps to understand emotions in text
- ✓ Useful for reviews, feedback, and social media analysis
- ✓ Simple yet powerful metric for sentiment analysis

Use Cases



Customer Feedback



Product Reviews



Social Media Monitoring



Brand Reputation



Together, **Bag of Words** helps machines understand the **content**, and **Polarity Score** helps them understand the **sentiment**!

